

JKBOSE 2026

CLASS 11TH

**LIMITS AND
DERIVATIVES**

MATHS

LECTURE-1

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TOPICS TO BE COVERED

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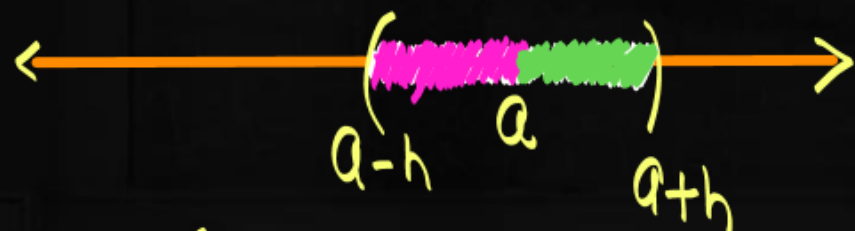
Introduction to Limits

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Related Problems

INTRODUCTION

Neighbourhood of a point: Suppose that $x=a$ be any real number, then the open interval $(a-h, a+h)$, where $h>0$ is said to be the neighbourhood of a point a .



* The Interval $(a-h, a]$ is said to be the left nbd. of a .

* The Interval $[a, a+h)$ is said to be the right nbd. of a .

★ **Limit of a function at a point:**

Question: Suppose that $f(x) = x^2$ be the given function, that what happens to $f(x)$ as x approaches to 2?



(Sol) At $x = 1.9$, $f(x) = (1.9)^2 = 3.61$

At $x = 1.99$, $f(x) = (1.99)^2 = 3.9601$

At $x = 1.999$, $f(x) = (1.999)^2 = 3.996001$

This shows that $f(x)$ approaches to 4 as x approaches to 2 in the left nbd. of 2.

i.e $f(x) \longrightarrow 4$ as $x \longrightarrow 2^-$

or $\lim_{x \rightarrow 2^-} f(x) = 4$ (left hand limit)

$$\text{At } x = 2.1, f(x) = (2.1)^2 = 4.41$$

$$\text{At } x = 2.01, f(x) = (2.01)^2 = 4.0401$$

$$\text{At } x = 2.001, f(x) = (2.001)^2 = 4.004001$$

This shows that $f(x)$ approaches to 4 as x approaches to 2 in the right nbd. of 2.

$$\text{i.e. } f(x) \longrightarrow 4 \text{ as } x \longrightarrow 2^+$$

$$\text{or } \lim_{x \rightarrow 2^+} f(x) = 4 \text{ (Right hand limit)}$$

Based on the above observation, we can conclude that

$f(x)$ approaches to 4 as x approaches to 2

$$\text{i.e. } f(x) \longrightarrow 4 \text{ as } x \longrightarrow 2$$

$$\text{or } \lim_{x \rightarrow 2} f(x) = 4 \text{ (limit of the function at } x=2)$$

Question: Suppose that $f(x) = \begin{cases} x+1; & x \geq 0 \\ x; & x < 0 \end{cases}$, then

What happens to $f(x)$ as x approaches to 0?



$$\text{(Sol)} \text{ At } x = -0.1, f(x) = -0.1$$

$$\text{At } x = -0.01, f(x) = -0.01$$

$$\text{At } x = -0.001, f(x) = -0.001$$

This shows that $f(x)$ approaches to 0 as x approaches to 0 in the left nbd. of 0.

$$\text{i.e. } f(x) \longrightarrow 0 \text{ as } x \longrightarrow 0^-$$

$$\text{or } \lim_{x \rightarrow 0^-} f(x) = 0 \text{ (left hand limit)}$$

$$\text{At } x = 0.1, f(x) = 0.1 + 1 = 1.1$$

$$\text{At } x = 0.01, f(x) = 0.01 + 1 = 1.01$$

$$\text{At } x = 0.001, f(x) = 0.001 + 1 = 1.001$$

This shows that $f(x)$ approaches to 1 as x approaches to 0 in the right nbd. of 0.

$$\text{i.e. } f(x) \rightarrow 1 \text{ as } x \rightarrow 0^+$$

$$\text{or } \lim_{x \rightarrow 0^+} f(x) = 1 \text{ (Right hand limit)}$$

Based on the above observation, we can conclude that $f(x)$ does not approach to a unique value as x approaches to 0. Hence the function has no limit at $x = 0$.

$$* \lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x) \longrightarrow \text{limit of } f \text{ exists at } x = a.$$

$$* \lim_{x \rightarrow a^+} f(x) \neq \lim_{x \rightarrow a^-} f(x) \longrightarrow \text{limit of } f \text{ does not exist at } x = a.$$

★ **Definition:** A real number l is said to be the limit of the function f at $x = a$ if $f(x)$ approaches to l as x approaches to a .

$$\text{i.e. } f(x) \rightarrow l \text{ as } x \rightarrow a \begin{pmatrix} x \rightarrow a^+ \\ x \rightarrow a^- \end{pmatrix}$$

Symbolically, we write it as;

$$\lim_{x \rightarrow a} f(x) = l.$$

Question: Suppose that $f(x) = \frac{x^2 - 25}{x - 5}$, $x \neq 5$ be the

given function, then what happens to $f(x)$ as x approaches to 5.

(Sol) Given function is;

$$f(x) = \frac{x^2 - 25}{x - 5} = \frac{x^2 - 5^2}{x - 5}, x \neq 5.$$

$$\Rightarrow f(x) = \frac{(x+5)(\cancel{x-5})}{(\cancel{x-5})}, x \neq 5.$$

$$\Rightarrow \boxed{f(x) = x + 5, x \neq 5}$$



At $x = 4.9$, $f(x) = 4.9 + 5 = 9.9$.

At $x = 4.99$, $f(x) = 4.99 + 5 = 9.99$

At $x = 4.999$, $f(x) = 4.999 + 5 = 9.999$.

This shows that $f(x)$ approaches to 10 as x approaches to 5 in the left nbd. of 5.

i.e. $f(x) \rightarrow 10$ as $x \rightarrow 5^-$

or $\lim_{x \rightarrow 5^-} f(x) = 10$ (left hand limit).

$$\text{At } x = 5.1, f(x) = 5.1 + 5 = 10.1.$$

$$\text{At } x = 5.01, f(x) = 5.01 + 5 = 10.01.$$

$$\text{At } x = 5.001, f(x) = 5.001 + 5 = 10.001.$$

This shows that $f(x)$ approaches 10 as x approaches 5 in the right nbd. of 5.

$$\text{i.e. } f(x) \longrightarrow 10 \text{ as } x \longrightarrow 5^+.$$

$$\text{or } \lim_{x \rightarrow 5^+} f(x) = 10 \text{ (Right hand limit)}$$

Based on the above observation, we can

Conclude that $f(x)$ approaches 10 as x approaches 5.

$$\text{i.e. } f(x) \longrightarrow 10 \text{ as } x \longrightarrow 5$$

$$\text{or } \lim_{x \rightarrow 5} f(x) = 10$$

$$\Rightarrow \lim_{x \rightarrow 5} \left(\frac{x^2 - 25}{x - 5} \right) = 10.$$

$$* \lim_{x \rightarrow a^+} f(x) = \lim_{h \rightarrow 0} f(a+h); \quad x = a+h.$$

$$* \lim_{x \rightarrow a^-} f(x) = \lim_{h \rightarrow 0} f(a-h); \quad x = a-h.$$

QUESTION

If $\lim_{x \rightarrow a} f(x)$ exists, then $f(a)$ must exist.

(A) True ✗

(B) False ✓✓

QUESTION

If $\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$, then the limit at $x = a$ does not exist.

- (A) True ✓
- (B) False ✗

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THANK

You